

3. wzory skróconego mnożenia 3

Zapisz w postaci sumy algebraicznej:

a) $(x-1)(x^2+x+1)(x^3+1)$

b) $(x+2)(x^4+4x^2+16)(x-2)$

$$\begin{aligned} \text{a)} \quad & \underbrace{(x-1)(x^2+x+1)}_{a^3-b^3=(a-b)(a^2+ab+b^2)} \underbrace{(x^3+1)}_{a^2-b^2=(a-b)(a+b)} = \underbrace{(x^3-1)(x^3+1)}_{=} = \\ & = (x^3)^2 - 1^2 = x^6 - 1 \end{aligned}$$

Odp: $x^6 - 1$.

$$\begin{aligned} \text{b)} \quad & \underbrace{(x+2)(x-2)}_{a^2-b^2=(a-b)(a+b)} (x^4+4x^2+16) = \underbrace{(x^2-4)(x^4+4x^2+16)}_{=} = \\ & = (x^2-4)(x^4+4x^2+16) = (x^2)^3 - (4)^3 = x^6 - 64 \end{aligned}$$

Odp: $x^6 - 64$.